# Sina Abdipoor (思纳), Ph.D.

AI Engineer, Researcher, and Teacher specializing in Operations Research, Optimization, and Machine Learning

Kuala Lumpur, Malaysia | sinaa@sunway.edu.my | sinaabdipoor.github.io | sina-abdipoor
SinaAbdipoor | Google Scholar | Sina-Abdipoor | 0000-0001-8411-5451 | Full CV | Scopus
All publications

## Summary

I'm a Lecturer in the Department of Data Science and Artificial Intelligence at Sunway University, Malaysia. I use Operations Research, Optimization, and Machine Learning to solve real-world problems, especially scheduling and timetabling. I hold a Ph.D. in Computer Science from Universiti Putra Malaysia.

My research focuses on metaheuristics and hybrid algorithms for combinatorial and stochastic optimization, and on judging algorithms by how well they generalize rather than how well they fit one benchmark. I have published in IEEE Access, Intelligent Systems with Applications, and GECCO. I collaborate with researchers in Malaysia and China, and I review for Expert Systems with Applications, Applied Soft Computing, and IEEE Transactions on Intelligent Transportation Systems.

From 2020 to 2025, I also worked as an independent Operations Research consultant. I helped organizations find inefficiencies, built optimization and prediction models, and presented the results to management.

I have taught since 2016, first English and later computer science and AI. Today I teach machine learning and software engineering, and I supervise Master's and undergraduate research.

## Research Interests

**Research:** Artificial Intelligence, Operations Research, Combinatorial Optimization, Metaheuristics, Evolutionary Computing, Scheduling and Timetabling, Machine Learning

**Teaching:** Machine Learning, Artificial Intelligence, Optimization and Search Algorithms, Evolutionary Computing, Programming, Software Engineering

**SDGs:** SDG 9 (Industry, Innovation, and Infrastructure), SDG 11 (Sustainable Cities and Communities), SDG 12 (Responsible Consumption and Production)

## Experience

**Sunway University**, Lecturer, Department of Data Science and Artificial Intelligence — Dec 2025 – present

- Teaching postgraduate and undergraduate courses in machine learning, software engineering, and data science capstone projects.
- Supervising 2 Master's students and more than 20 undergraduate final year projects.
- Researching metaheuristics and hybrid optimization for real-world scheduling and timetabling, with 8 papers published in 2026.
- Collaborating with Hefei University, China, on timetabling and evolutionary computation, with joint papers at GECCO 2026 and PATAT 2026.
- Contributing to program development, quality assurance, and department committees.

**Universiti Putra Malaysia**, Graduate Researcher & Tutor — Sept 2020 – Sept 2025

- Researched hybrid metaheuristics for university course timetabling for my Ph.D., published in IEEE Access and Intelligent Systems with Applications.
- Taught Computer Programming 2 (SSK3101) and Machine Learning (SSK5603) at undergraduate and postgraduate levels.

**Independent Consultant**, Operations Research Engineer — Jan 2020 – Mar 2025

- Partnered with organizations to analyze workflows and find inefficiencies in resource allocation.
- Designed, proposed, and deployed optimization models.
- Presented findings and recommendations to stakeholders and management.

**Putra Graduate Association, Universiti Putra Malaysia**, Chief Exco of International Relations Affairs & Alumni — July 2022 – July 2023

- Organized programs for international postgraduate students and alumni.
- Coordinated postgraduate clubs and student country representatives, in Malaysia and abroad.
- Maintained the UPM postgraduate alumni records.
- Represented the association at iPutra and Education Malaysia Global Services (EMGS).

**Pars Six Sigma Excellence**, Software Engineer — July 2017 – July 2019

- Designed, built, and maintained a CRM system for employees and customers in a small team of software engineers.
- Developed the Java backend using a variety of libraries, frameworks, and APIs.
- Designed and implemented the SQL database.

**Kish Institute of Science and Technology**, Senior English Teacher & Supervisor — Nov 2016 – May 2019

- Taught Upper-Intermediate and Advanced English (B2 to C2) to teenagers and adults, following Oxford's syllabus.
- Led IELTS workshops, level assessments, and teacher interviews.
- Observed classes to keep teaching standards high.

**IKCO**, Android Developer (Internship) — June 2016 – Sept 2016

- Built an Android app with SQLite that lets assembly line workers check parts stock in real time and order parts without leaving their station.

**ParsOnline**, Network Engineer — June 2014 – Mar 2015

- Installed and configured network equipment.

## Education

**Ph.D.** **Universiti Putra Malaysia**, Computer Science — 2020 – 2026
Thesis: Optimality and Generality of Metaheuristics in University Course Timetabling

- Proposed a hybrid metaheuristic for University Course Timetabling with Student Sectioning (UCTSSP), tested on the International Timetabling Competition 2019 benchmark.
- Supervisor: Associate Professor Dr. Razali Yaakob. Committee: Professor Dr. Salwani Abdullah, Dr. Khairul Azhar Kasmiran, Dr. Hazlina Hamdan
- Collaborators: Associate Professor Dr. Say Leng Goh, Professor Graham Kendall

**M.Sc.** **K. N. Toosi University of Technology**, Computer Engineering, Artificial Intelligence and Robotics — 2016 – 2019
Thesis: Analyzing Parameter Tuning of Evolutionary Algorithms in Optimization Problems

- Supervisor: Associate Professor Dr. Amin Nikanjam
- Courses: Machine Learning, Reinforcement Learning, Evolutionary Computing, Pattern Recognition, Probabilistic Graphical Models, Artificial Neural Networks, Big Data

**B.Sc.** **Imam Khomeini International University**, Computer Engineering, Software Engineering — 2012 – 2016

- Final year project: Designing, Implementing, and Launching a Restaurant Management System for a Local Cafe
- Supervisor: Associate Professor Dr. Morteza Mohammadi Zanjireh
- Courses: Artificial Intelligence, Data Mining, Algorithm Design, Database Design, Object-Oriented Programming, Data Structures

## Publications

**An integration of discrete wavelet transform and time-series recurrent neural network to improve stock price prediction** — Aug 2026
Keong Kuon Lock, Razali Yaakob, **Sina Abdipoor**, Mingxuan Li
10.1108/ACI-08-2025-0335 (Applied Computing and Informatics)

**Integrative Hybrid Local Search with Luby's Restart Strategy for University Course Timetabling with Student Sectioning** Aug 2026
Mingxuan Li, Thomas Weise, **Sina Abdipoor**, Jourdan D'orville, Say Leng Goh, Razali Yaakob, Salwani Abdullah
thomasweise.github.io/research/publications (15th Conference on the Practice and Theory of Automated Timetabling (PATAT))

**SpOC 4 Challenges 1 Beginner: Permutation Encodings and Challenge 3: Simulated Annealing for the Luna Tomato Advertising** July 2026
Mingxuan Li, Thomas Weise, **Sina Abdipoor**, Zhi-Ze Wu
10.1145/3795101.3815558 (Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO))

**Solving the SpOC 4 Challenge 2 with a Multi-Step Permutation Encoding for the Keplerian Traveling Salesperson** July 2026
Thomas Weise, **Sina Abdipoor**, Mingxuan Li, Zhi-Ze Wu
10.1145/3795101.3815557 (Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO))

**A Machine Learning Approach to Forecast Car Valuation in the Automotive Industry** June 2026
Chun-Yun Chiang, **Sina Abdipoor**, Anwar P. P. Abdul Majeed, D. O. Alebiosu, Wan Siti Nur Aiza, Muhammed Basheer Jasser
10.1109/I2CACIS69435.2026.11600322 (2026 IEEE International Conference on Automatic Control and Intelligent Systems (I2CACIS))

**A Review on Artificial Intelligence Techniques Employed for Enhancing NPC Behavior in Video Games** June 2026
Ruben Raj Karthigasu, Muhammed Basheer Jasser, Samuel Mofoluwa Ajibade, **Sina Abdipoor**, Na-Chua Hui
10.1109/I2CACIS69435.2026.11600010 (2026 IEEE International Conference on Automatic Control and Intelligent Systems (I2CACIS))

**Real-World University Timetabling: A Two-Stage Collaborative Hybrid Metaheuristic** May 2026
Chee Seng Leong, Say Leng Goh, **Sina Abdipoor**, Muhammed Basheer Jasser, Na-Chua Hui, Hao Luo
10.1109/CSPA68262.2026.11517821 (2026 IEEE International Colloquium on Signal Processing & Its Applications (CSPA))

**A Comparison Study of Swarm Intelligence Algorithms Employed for Optimizing ANN Training in Flood Prediction** May 2026
Bibi Aamirah Shafaa Emambocus, Muhammed Basheer Jasser, Samuel Mofoluwa Ajibade, **Sina Abdipoor**, Na-Chua Hui
10.1109/CSPA68262.2026.11517716 (2026 IEEE International Colloquium on Signal Processing & Its Applications (CSPA))

**Urban House Price Forecasting in Malaysia Using Tree-Based Machine Learning Models** Apr 2026
Ying Zheng, Hui Shen Choo, Yoke Leng Yong, Umi Najiah Ahmad Razimi, Muhammed Basheer Jasser, **Sina Abdipoor**
10.1109/Robothia68364.2026.11507118 (2026 IEEE 2nd International Conference on Robotics and Technologies for Industrial Automation (ROBOTHIA))

**Distance to Feasibility: Performance Assessment of Baseline Metaheuristics on University Course Timetabling with Student Sectioning** Dec 2025
**Sina Abdipoor**, Mingxuan Li, Say Leng Goh, Razali Yaakob, Hao Luo, Salwani Abdullah
10.1109/ICSPC68261.2025.11325835 (2025 13th IEEE Conference on Systems, Process & Control (ICSPC))

**Optimality Versus Generality: Performance Assessment of Meta-Heuristics in Educational Timetabling** 2025
**Sina Abdipoor**, Razali Yaakob, Say Leng Goh, Salwani Abdullah, Hazlina Hamdan, Khairul Azhar Kasmiran
10.1109/ACCESS.2025.3565522 (IEEE Access)

**Digital Twin System for Smart Campuses** Oct 2024
Hui-Jie Wang, Yuanyuan Cai, Fabu Yue, Hao Luo, **Sina Abdipoor**, Say Leng Goh
10.1109/ICCSSE63803.2024.10823754 (2024 IEEE 10th International Conference on Control Science and Systems Engineering (ICCSSE))

**Hybrid Graph Neural Networks with LSTM Attention Mechanism for Recommendation Systems in MOOCs** Mar 2024
Best Paper Award
Hao Luo, **Sina Abdipoor**, Qing Li, Li-Tao Xu
10.1109/CSPA60979.2024.10525319 (2024 20th IEEE International Colloquium on Signal Processing & Its Applications (CSPA))

**Smart-Kube: Energy-Aware and Fair Kubernetes Job Scheduler Using Deep Reinforcement Learning** Sept 2023
Saeid Ghafouri, **Sina Abdipoor**, Joseph Doyle
10.1109/SmartCloud58862.2023.00035 (2023 IEEE 8th International Conference on Smart Cloud (SmartCloud))

**Meta-heuristic approaches for the University Course Timetabling Problem** Sept 2023
**Sina Abdipoor**, Razali Yaakob, Say Leng Goh, Salwani Abdullah
10.1016/j.iswa.2023.200253 (Intelligent Systems with Applications)

**International Timetabling Competition 2019: A Systematic Literature Review** June 2023
**Sina Abdipoor**, Razali Yaakob, Say Leng Goh, Salwani Abdullah, Khairul Azhar Kasmiran, Hazlina Hamdan
10.1109/I2CACIS57635.2023.10193383 (2023 IEEE International Conference on Automatic Control and Intelligent Systems (I2CACIS))

## Awards

**Best Paper Award** Aug 2026
19th International Conference on Advanced Computer Theory and Engineering (ICACTE)

**National Finalist, 3 Minute Thesis (3MT) Competition** June 2024
Universiti Kebangsaan Malaysia

**Winner, 3 Minute Thesis (3MT) Competition** Mar 2024
Universiti Putra Malaysia

**Best Paper Award** Mar 2024
20th IEEE International Colloquium on Signal Processing & Its Applications (CSPA)

**Special Jury Award, Postgraduate Colloquium** Oct 2023
Universiti Putra Malaysia

**Special Graduate Research Allowance (SGRA)** June 2021
Universiti Putra Malaysia, grant FRGS/1/2020/ICT02/UPM/02/5

**Rank 18, National University Entrance Exam (Konkoor), Private Universities, Computer Engineering** May 2016
National Organization for Educational Testing (Sanjesh)

**Rank 179, National University Entrance Exam (Konkoor), Governmental Universities, Computer Engineering** May 2016
National Organization for Educational Testing (Sanjesh)

**Silver Medal, National Universities Swimming Championship** 2013
Imam Khomeini International University

## Talks

**Applied Operations Research in Driving Sustainability** — China
Guest talk, Hefei University — Aug 2026

**AI in Higher Education and AI Governance** — May 2026
Invited interview, University of Lincoln

**Preparing New UPM Ph.D. Students for 3MT 2025** — Malaysia
Sharing session, Universiti Putra Malaysia — 2025

## Conference Presentations

**19th International Conference on Advanced Computer Theory and Engineering (ICACTE 2026)** — China 2026

**15th Conference on the Practice and Theory of Automated Timetabling (PATAT 2026)** — United Kingdom 2026

**67th Annual Conference of the Canadian Operational Research Society (CORS)** — Canada 2026

**Genetic and Evolutionary Computation Conference (GECCO 2026)** — Costa Rica 2026

**IEEE International Conference on Automatic Control & Intelligent Systems (I2CACIS)** — Malaysia 2026

**22nd IEEE International Colloquium on Signal Processing & Its Applications (CSPA)** — Malaysia 2026

**2nd IEEE International Conference on Robotics and Technologies for Industrial Automation (ROBOTHIA)** — Malaysia 2026

**13th IEEE Conference on Systems, Process & Control (ICSPC)** — Malaysia 2025

**66th Annual Conference of the Canadian Operational Research Society (CORS)** — Canada 2025

**10th IEEE International Conference on Control Science and Systems Engineering (ICCSSE)** — China 2024

**65th Annual Conference of the Canadian Operational Research Society (CORS)** — Canada 2024

**20th IEEE International Colloquium on Signal Processing & Its Applications (CSPA)** — Malaysia 2024

**8th IEEE International Conference on Smart Cloud (SmartCloud)** — Japan 2023

**IEEE International Conference on Automatic Control & Intelligent Systems (I2CACIS)** — Malaysia 2023

## Teaching

**Applied Machine Learning (MAI5043)** — Feb 2026
Lecturer, Master in Artificial Intelligence, Sunway University

**Software Engineering (SEG2202)** — Feb 2026
Lecturer, B.Sc. Computer Science, Sunway University

**Capstone Project 1 and 2 (MRP5015, MRP5025)** — Mar 2026
Lecturer, Master in Data Science (ODL), Sunway University

**Machine Learning (SSK5603)** 2020
Tutor, Universiti Putra Malaysia

**Computer Programming 2 (SSK3101)** 2020
Tutor, Universiti Putra Malaysia

**Evolutionary Computing** 2018
Teaching Assistant, K. N. Toosi University of Technology

**Algorithm Design** 2014
Teaching Assistant, Imam Khomeini International University

## Supervision

**Whong Ye Wen** 2026 – present
Master in Data Science, Sunway University, Main Supervisor

**Tan Yi Wei** 2026 – present
Master in Artificial Intelligence, Sunway University, Main Supervisor

**More than 20 undergraduate final year projects** 2026 – present
Sunway University

**Mingxuan Li** 2025 – present
B.Sc., Queen Mary University of London, Main Supervisor

## Peer Review

**Applied Soft Computing, Elsevier** 2026

**IEEE Transactions on Intelligent Transportation Systems** 2026

**Expert Systems with Applications, Elsevier** 2025

**Intelligent Systems with Applications, Elsevier** 2023 – 2025

**IEEE Access** 2024

**International Journal of Artificial Intelligence** 2024

## Professional Memberships

**Canadian Operational Research Society (CORS)** Apr 2024 – present
Member

## Projects

**ESA Space Optimisation Competition (SpOC 4)** 2026
Ranked 4th globally in one track of the European Space Agency's space logistics optimization challenge at GECCO 2026, using permutation encodings and simulated annealing. Our solutions were published as two GECCO 2026 papers.

**UPM Thesis LaTeX Template** 2024
A LaTeX thesis template for Universiti Putra Malaysia, built from the official Microsoft Word template.

**International Timetabling Competition 2019** 2022
An implementation of the International Timetabling Competition 2019 benchmark, the latest benchmark for university course timetabling, built as the core of my Ph.D.

**Car Evaluation Dataset** 2021
A study of feature engineering and of how different classification methods perform on the Car Evaluation dataset.

**Java Framework for Genetic Algorithms** 2019
A Java framework to implement evolutionary algorithms, apply them to different problems, and compare their performance. Built during my master's research.

**Tic-Tac-Toe Game** 2018
A game with an AI opponent that uses Q-Learning and has several difficulty levels. Built for a Reinforcement Learning course.

**Photo Noise Reduction** 2018
Reduces image noise with a message passing technique. Final project for a Probabilistic Graphical Models course.

**Weather and Traffic Forecasting Software** 2017
Forecasts weather and traffic density with a neural network trained on historical data. Built for an Artificial Neural Networks course.

**Auto Parts Finder** 2016
An Android app used by assembly line workers at IKCO to check parts stock in real time and order parts, which cut idle time on the line.

## Skills

**Operations Research:** Combinatorial Optimization, Mixed Integer Programming, Stochastic Optimization, Constraint Programming, Metaheuristics, Linear and Mathematical Programming, Gurobi, Google OR-Tools

**Data Science:** Predictive Modeling, Supervised and Unsupervised Learning, Data Visualization, TensorFlow, Scikit-Learn, NumPy, Pandas, Matplotlib, Power BI, Tableau, Excel

**Programming:** Python, Java, SQL, MySQL, SQLite, XML, Git, GitHub, UML

**Research and Teaching:** Scientific Writing, LaTeX, Public Speaking, Presenting, Teaching

**Languages:** English (Bilingual, IELTS 8), Mandarin (Intermediate), Malay (Basic)

## Certificates

**Supervisory Training Certificate (SURE)** 2026

**Full Course Curriculum Design Workshop and Certificates for MOHE and CQM** 2026
OBE, Table 4, ODL, and rubrics

**Chinese (Mandarin) Certificate** May 2021
Peking University

**Google Data Analytics Certificate** Apr 2021
Google

**IELTS Academic, Overall Band 8** June 2019
IELTS

**Certificate in Advanced English (CAE), C2** Oct 2016
Cambridge Assessment English

**Cambridge Teacher Training Course (TTC)** July 2016
Kish Institute of Science and Technology

## Research Projects

**Performance Analysis of Classification Methods in Predicting Chinese International Students' Decision to Return Home due to the COVID-19 Outbreak** 2021

**A Model for Evaluating the Impact of Incrementing Size of Population on the Performance of Evolutionary Algorithms** 2020

## Workshops and Seminars

**Classifying Problems by Computational Complexity: P vs NP and Beyond, by Professor Paul Goldberg** 2025

**CORS Micro-event on Hybrid Quantum-Classical Computing for Internet of Everything Optimization, by Dr. Mahzabeen Emu** 2025

**Best Practices of Teaching Supply Chain Analytics to Meet Job Market Requirements** 2025

| | |
|---|---|
| **Deutscher Akademischer Austauschdienst (DAAD) Webinar** | 2025 |
| **Google I/O** | 2025 |
| **CORS Transportation & Logistics SIG Seminar: The Future of Mobility, From Gridlock to Green Solutions, by Leandro Coelho** | 2025 |
| **CORS Micro-event on Public Transit and Supply Chains: How Data Can Support Optimization in Public Transit Operations, by Professor Martin Trepanier** | 2025 |
| **CORS Micro-event on Analytics: An OR Tale of Analytics, AI, and Auctions, by Professor Fredrik Ødegaard** | 2024 |
| **How to (Ethically) Increase Your Citations, by Professor Graham Kendall** | 2024 |
| **CORS SIG in Energy, Environment, and Sustainability: Optimization Challenges in Electric Vehicle Charging Operations, by Professor Miguel Anjos** | 2024 |
| **Expert Talk: Graph-based Explainable and Fair Recommendation, by Professor Guandong Xu** | 2024 |
| **2nd CORS Micro-Event on Circular Economy and Autonomous Mobility** | 2024 |
| **Tips and Guide for Publishing in High-Impact Journals** | 2024 |
| **Elsevier Workshop on Research Data Management: Dive Into the Details** | 2024 |
| **AI Tools for Literature Review Search and Writing** | 2024 |
| **Online Database Classes: Scopus** | 2024 |
| **United States Air Force Office of Scientific Research (AFOSR) International Grant Funding Briefing** | 2024 |
| **NVIDIA Research Workshop** | 2023 |
| **Postgraduate Colloquium, Universiti Putra Malaysia** | 2023 |
| **Research Showcase at FSKTM Open Day, Universiti Putra Malaysia** | 2023 |
| **Elsevier Young Researchers Workshop** | 2023 |
| **UPM Workshop on High-Impact Publishing** | 2022 |
| **IEEE Young Professionals Malaysia Seminar** | 2021 |
| **Workshop on How to Publish with Oxford Journals** | 2021 |

## References

**Associate Professor Dr. Razali Yaakob**
Universiti Putra Malaysia, razaliy@upm.edu.my

**Professor Salwani Abdullah**
Universiti Kebangsaan Malaysia, salwani@ukm.edu.my

**Professor Dr. Thomas Weise**
Hefei University, tweise@hfuu.edu.cn

**Professor Anwar P. P. Abdul Majeed**
Head of Department of Data Science and Artificial Intelligence, Sunway University, anwarm@sunway.edu.my

**Professor Graham Kendall**
University of Nottingham, graham.kendall@nottingham.ac.uk